

UNIVERSITÄT TRIER
BUSINESS ANALYTICS

Descriptive Insights and Predictive Customer Value

An Empirical Analysis of Amazon Purchase Patterns

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Abstract

This document reports Part A and Part B as two related but separate pieces of work using the published Open e-commerce 1.0 dataset, which links crowdsourced U.S. Amazon purchase histories to survey-based customer characteristics¹². For the primary analysis, we use complete observations from January 2018 through December 2022. Part A is a descriptive analysis of customers, products, places, and time periods associated with observed value in the sample. Part B is a predictive analysis that tests whether behaviour observed from 2018 can identify customers who become high value in the future.

Part A establishes a strongly right-skewed transaction-value distribution: the mean line-item value is \$23.69, the median is \$14.49, the conventional IQR outlier threshold is \$48.82, and the 99th percentile is \$180.46. Books generated the highest revenue among original categories (\$1.36 million), while Groceries & Food was the most frequently purchased interpretable aggregated category. The \$100,000-\$149,999 income segment, customers aged 25-34, California, and Mondays were the highest-revenue groups in their respective analyses. In the state-based analysis, California is the leading observed state. These are sample-level associations, not population estimates or causal effects³.

Part B converts the descriptive evidence into a binary classification design. Each eligible customer is represented by features calculated exclusively from 2018-2020. Future spending in 2021-2022 defines the target: customers at or above \$5,908.16, the 75th percentile, are labelled high value. The final customer-level dataset contains 4,882 customers, including 1,221 high-value customers (25.01%). An 80/20 stratified split leaves 977 customers for final testing⁴.

Among the untuned models, Gradient Boosting performed best with ROC-AUC 0.919 and F1-score 0.748. After cross-validated tuning, its ROC-AUC was 0.920 on the untouched test set. At a 0.50 threshold, the tuned model correctly classified 679 non-high-value and 175 high-value customers, with 54 false positives and 69 false negatives. The high-value class achieved precision 0.76, recall 0.72, and F1-score 0.74. Behavioural features were substantially more predictive than demographics alone, although the combined feature set performed best.

Key managerial conclusion

Amazon should use behavioural signals, especially historical monetary value, category diversity, purchasing intensity, and recency to prioritize retention, cross-selling, and re-engagement. Demographics should be treated as supplementary context and monitored for fairness, not as the sole basis for targeting. Model probabilities should support resource allocation, while campaign thresholds must be selected according to cost, capacity, and the relative consequences of false positives and false negatives.

¹ Berke, A., et al. (2024). Open e-commerce 1.0: Five years of crowdsourced U.S. Amazon purchase histories with user demographics. Scientific Data. <https://doi.org/10.1038/s41597-024-03329-6>

² Harvard Dataverse. Open e-commerce 1.0 dataset. <https://doi.org/10.7910/DVN/YGLYDY>

³ Wibisono, I. A., Agarwal, B., & Schander, R. (2026). *Project A* [Google Colab notebook]. Google Colab. https://colab.research.google.com/drive/1LYEbNHwc8IXNQG5J9GoL0Vv_dTkzWcAc?usp=sharing

⁴ Wibisono, I. A., Agarwal, B., & Schander, R. (2026). *Project B* [Google Colab notebook]. Google Colab. <https://colab.research.google.com/drive/1atpBp7PEz7Ntv10QSUHvwBV5ryDidrn-?usp=sharing>

Table of Contents

1. Project Context and Research Questions.....	4
1.1 Business context.....	4
1.2 Business Problems.....	4
1.3 Analytical structure.....	4
1.4 Scope of the document.....	4
2. Data, Integration, and Governance.....	5
2.1 Data sources and unit of analysis.....	5
2.2 Data preparation.....	5
2.3 Data quality decisions.....	5
2.4 Governance and privacy.....	6
3. Part A - Descriptive Analytics.....	7
3.1 Objective and method.....	7
3.2 Transaction value distribution.....	7
3.3 Product patterns.....	8
3.4 Geographic and demographic patterns.....	9
3.5 Temporal patterns.....	11
3.6 Part A business implications.....	11
4. Part B Research Design.....	12
4.1 Part B objective.....	12
4.2 Temporal prediction design.....	12
4.3 Target distribution.....	12
5. Part B - Predictive Analytics.....	13
5.1 Feature engineering.....	13
5.2 Preprocessing pipeline.....	13
5.3 Candidate models and evaluation.....	13
5.4 Untuned model comparison.....	14
5.5 Feature-set experiment.....	15
5.6 Hyperparameter tuning.....	16
5.7 Final test-set evaluation.....	17
5.8 Model interpretation.....	18
5.9 Business translation.....	19
5.10 Requirement coverage.....	19
6. Part B Business Recommendations.....	20
6.1 Recommended operating model.....	20
6.2 Campaign threshold framework.....	20
6.3 Measurement plan.....	20
6.4 Deployment sequence.....	20
7. Conclusion.....	21
8. Code Availability and Reproducibility.....	22
References.....	23
Appendix A - Detailed Part A Rankings.....	24
Appendix B - Predictive Feature Dictionary.....	25

1. Project Context and Research Questions

1.1 Business context

E-commerce platforms record large volumes of transaction-level data, yet business decisions are usually made at higher levels such as customer segments, product categories, regions, and campaign audiences. This document therefore reports two separate analytics tasks on the same dataset. Part A summarizes historical transaction and survey-linked patterns in the observed sample⁵. Part B then develops a predictive model for future high-value customer status using a temporally separated feature and outcome design⁶.

1.2 Business Problems

- **Part A:** How do age, income, education, geography, product category, and time relate to revenue and purchasing behaviour in the observed sample?
- **Part B:** How can Amazon use historical purchase behaviour and selected customer characteristics to identify customers with high future revenue potential and allocate marketing resources more efficiently?

1.3 Analytical structure

Stage	Analytical role	Primary output
Part A	Describe patterns in transactions and linked survey characteristics	Revenue distributions, segments, product/geographic/time patterns, dashboard
Part B	Predict future high-value status at customer level	Model comparison, tuned classifier, evaluation, interpretation, targeting strategy
Integration	Translate evidence into decisions while preserving limitations	Prioritized actions, governance controls, monitoring plan

1.4 Scope of the document

The source publication introduced and validated the dataset^{7 8}. This document uses the published records independently for business analytics coursework. The aim is not to propose a new algorithm, but to document a descriptive analysis in Part A and a predictive modelling workflow in Part B, each with clear methods, results, limitations, and business interpretation.

⁵ Wibisono, I. A., Agarwal, B., & Schander, R. (2026). *Project A* [Google Colab notebook]. Google Colab. https://colab.research.google.com/drive/1LYEbNHwc8IXNQG5J9GoL0Vv_dTkzWcAc?usp=sharing

⁶ Wibisono, I. A., Agarwal, B., & Schander, R. (2026). *Project B* [Google Colab notebook]. Google Colab. <https://colab.research.google.com/drive/1atpBp7PEz7Ntv10QSUHvwBV5ryDidrn-?usp=sharing>

⁷ Berke, A., et al. (2024). Open e-commerce 1.0: Five years of crowdsourced U.S. Amazon purchase histories with user demographics. Scientific Data. <https://doi.org/10.1038/s41597-024-03329-6>

⁸ Harvard Dataverse. Open e-commerce 1.0 dataset. <https://doi.org/10.7910/DVN/YGLYDY>

2. Data, Integration, and Governance

2.1 Data sources and unit of analysis

Source	Content	Role
amazon-purchases.csv	Order date, title, category, unit price, quantity, shipping state, Survey ResponseID	Transaction-level behavioural evidence
survey.csv	Age, income, education, gender, state, Amazon-use frequency and other responses	Selected customer context
Survey ResponseID	Shared anonymous identifier	Join key and customer unit for Part B

Part A operates mainly at transaction level and aggregates to categories, states, demographic groups, and time periods⁹. Part B aggregates historical purchases to one row per customer¹⁰. This distinction is important: transaction-level observations are not independent customer observations, and predictive splitting must occur after customer aggregation.

2.2 Data preparation

1. Parse order dates and derive year, month, and weekday.
2. Convert price and quantity to numeric values and compute Total Price = unit price x quantity.
3. Merge transactions with survey data on Survey ResponseID.
4. Standardize categorical variables and assign readable aggregated product groups.
5. Review missingness and preserve relevant demographic records using an Unknown label in Part B.
6. Validate that the predictive table contains exactly one row per customer.

The term purchase-record value is used throughout because the dataset does not provide a reliable basket-level checkout identifier.

2.3 Data quality decisions

The analysis retains the complete revenue distribution because large values are commercially meaningful. Visualizations are restricted where needed (for example, to the 99th percentile) so that typical purchases remain visible. The IQR rule is used to describe unusual transactions, not automatically to delete them. Product-category aggregation improves readability, but the residual Other / Misc group is not interpreted as a homogeneous business category.

The date-filtered dataset contained 1,810,793 purchase records. But for Part A analysis, 89,235 additional records with missing Category were excluded, resulting in 1,721,558 category-complete records.

⁹ Wibisono, I. A., Agarwal, B., & Schander, R. (2026). *Project A* [Google Colab notebook]. Google Colab. https://colab.research.google.com/drive/1LYEbNHwc8IXNQG5J9GoL0Vv_dTkzWcAc?usp=sharing

¹⁰ Wibisono, I. A., Agarwal, B., & Schander, R. (2026). *Project B* [Google Colab notebook]. Google Colab. <https://colab.research.google.com/drive/1atpBp7PEz7Ntv10QSUHvwBV5ryDidrn-?usp=sharing>

Year	Purchase records	Maximum date	Analytical decision
2018	225,465	31 Dec 2018	Included
2019	266,016	31 Dec 2019	Included
2020	409,212	31 Dec 2020	Included
2021	467,870	31 Dec 2021	Included
2022	442,230	31 Dec 2022	Included
2023	39,923	1 Sep 2023	Excluded: incomplete
2024	1	15 Aug 2024	Excluded: one record

Date-coverage audit and analytical-period decision

Although the raw file contains dates beyond 2022, the later coverage is incomplete. The core analysis therefore uses complete years from 2018 through 2022. This decision avoids interpreting partial-year records as full-year market trends.

2.4 Governance and privacy

Only the anonymous survey identifier is required for integration. The predictive model excludes *Future_Spending* from the input features and omits sensitive survey domains such as health, substance use, race, ethnicity, and sexual orientation. Age, income, gender, education, state, and stated Amazon-use frequency remain included as selected contextual features; their use requires fairness monitoring and purpose limitation.

3. Part A - Descriptive Analytics

3.1 Objective and method

Part A¹¹ uses summary statistics, frequency tables, grouped revenue aggregations, histograms, bar charts, time-series plots, a boxplot, and an interactive dashboard. The objective is to identify concentration, segment differences, and operational patterns before building a predictive model.

3.2 Transaction value distribution

Statistic	Observed value	Interpretation
Mean line-item value	\$23.69	Raised by relatively rare high-value purchases
Median line-item value	\$14.49	More representative of a typical line item
IQR outlier threshold	\$48.82	Descriptive flag; not an automatic deletion rule
99th percentile	\$180.46	Only about 1% of line items exceed this value

The mean exceeds the median, confirming a right-skewed distribution. Low-value line items dominate transaction volume, whereas a small tail of expensive purchases can contribute disproportionate revenue. Consequently, decisions based only on an average purchase value would obscure both the high-volume low-value base and the commercially important upper tail.

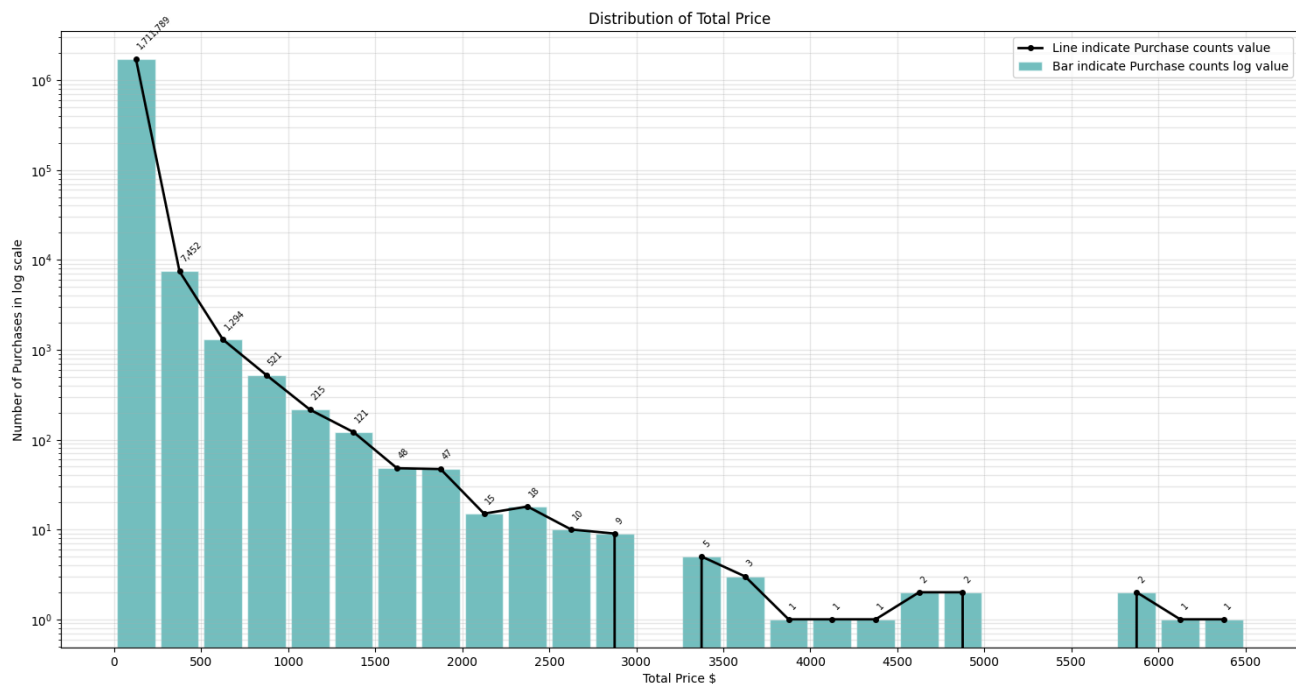


Figure 1 - Distribution of purchase-record value in the 2018-2022 analytical dataset

Figure 1 shows a strongly right-skewed distribution of purchase-record value across the full observed range. Most purchase records are low value, while a small number of high-value purchases extend the right tail to approximately \$6,400. The mean is higher than the median because a relatively small number of expensive purchases raise the average.

¹¹ Wibisono, I. A., Agarwal, B., & Schander, R. (2026). *Project A* [Google Colab notebook]. Google Colab. https://colab.research.google.com/drive/1LYEbNHwc8IXNQG5J9GoL0Vv_dTkzWcAc?usp=sharing

3.3 Product patterns

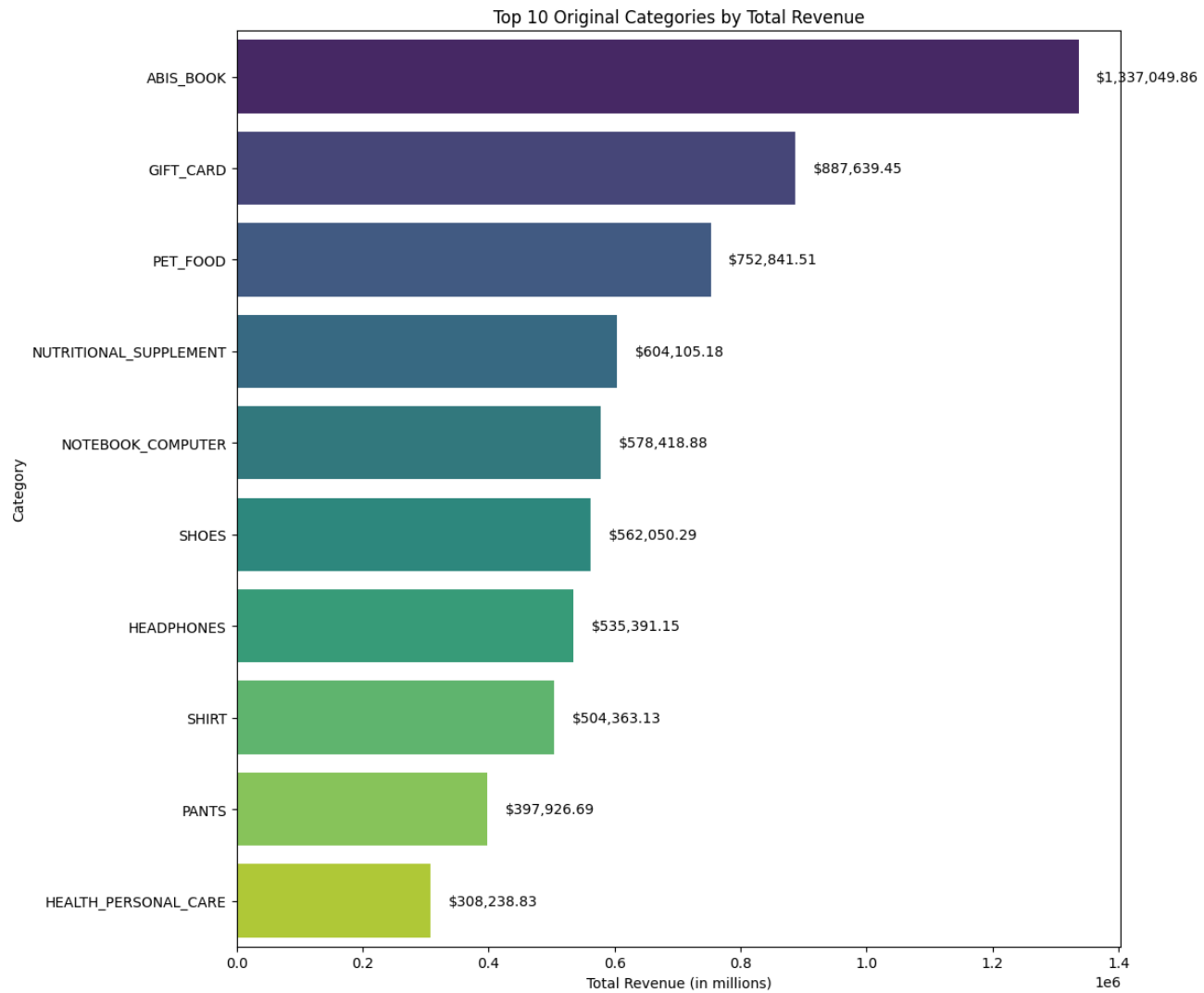


Figure 2 - Revenue of the ten leading original product categories in the analysed sample

ABIS_BOOK leads the original category ranking with approximately \$1.33 million, followed by gift cards (\$0.88 million) and pet food (\$0.75 million). At the aggregated-category level, Other / Misc is largest because it combines many heterogeneous residual categories. Excluding that artificial residual, Groceries & Food is the most frequently purchased aggregated category, with more than 200,000 purchases reported in the notebook analysis.

3.4 Geographic and demographic patterns

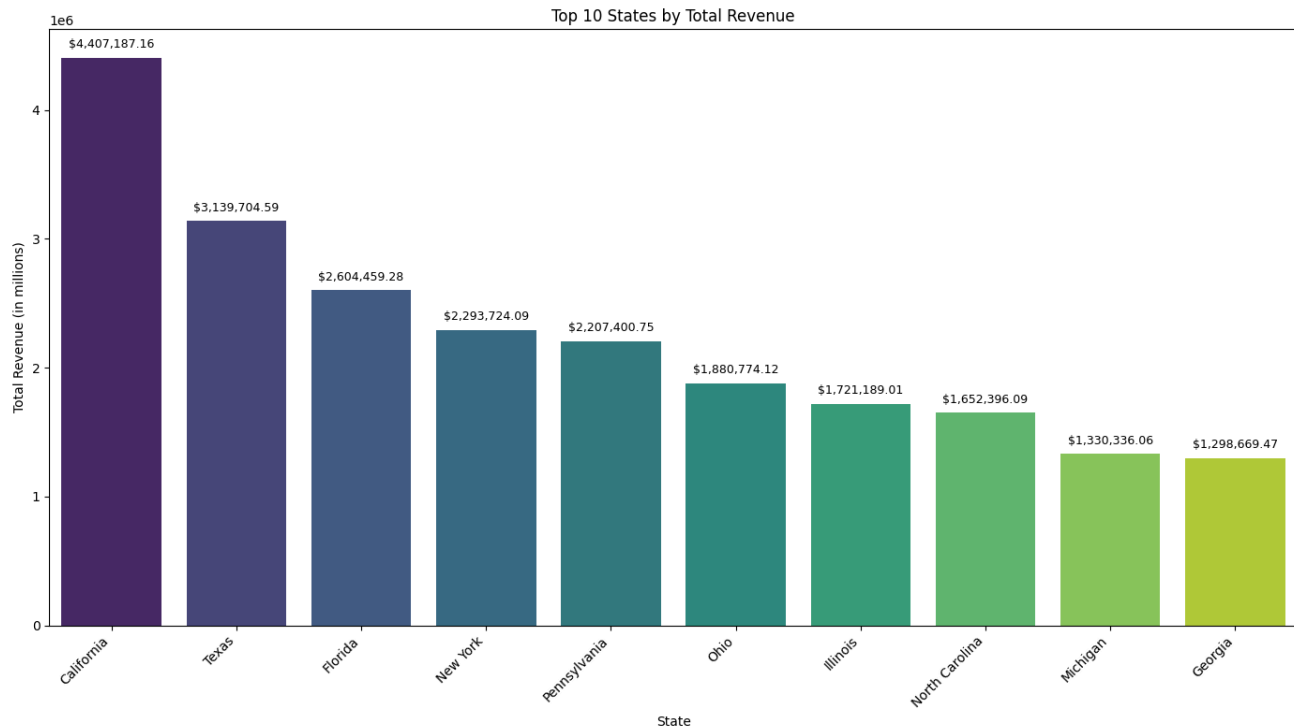


Figure 3 - Top ten states by revenue

Dimension	Highest observed group	Reported magnitude
State	California	\$4.4 million
Income	\$100,000-\$149,999	\$8.93 million
Age	25-34 years	More than \$13 million
Weekday	Monday	More than \$6 million

California generated the most revenue in the report's state-based analysis, followed by Texas and Florida. In Part A, this analysis is based on survey-reported customer state (Q-demos-state) rather than the sometimes-missing shipping-address state. The revenue ranking partly reflects the number of sampled users and market size, so it should not be read as spending propensity without normalization by customer count. Similarly, total revenue by age or income combines segment size and individual spending. These totals are useful for resource planning but should be complemented by per-customer measures before deciding that one demographic group is intrinsically more valuable.

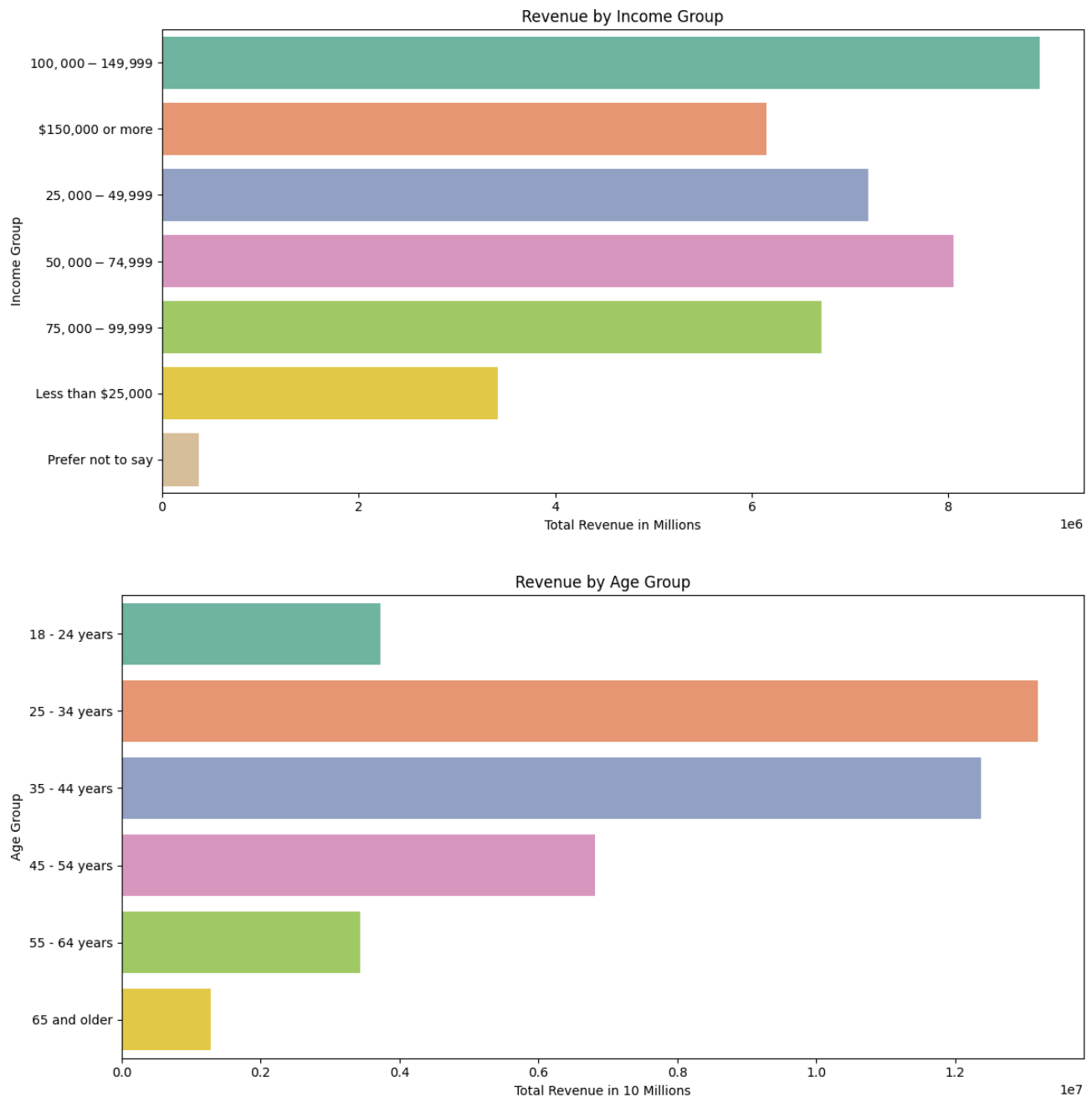


Figure 4 and 5 - Total purchase-record value by customer income and age group in the 2018–2022 analytical sample. Bars show aggregate segment value and therefore reflect both segment size and spending intensity.

Figure shows that aggregate purchase-record value varies across demographic groups. The \$100,000–\$149,999 income group contributes the highest observed total value, while customers aged 25–34 form the largest age-based revenue segment. They indicate each segment opportunity rather than evidence about individual spending.

3.5 Temporal patterns

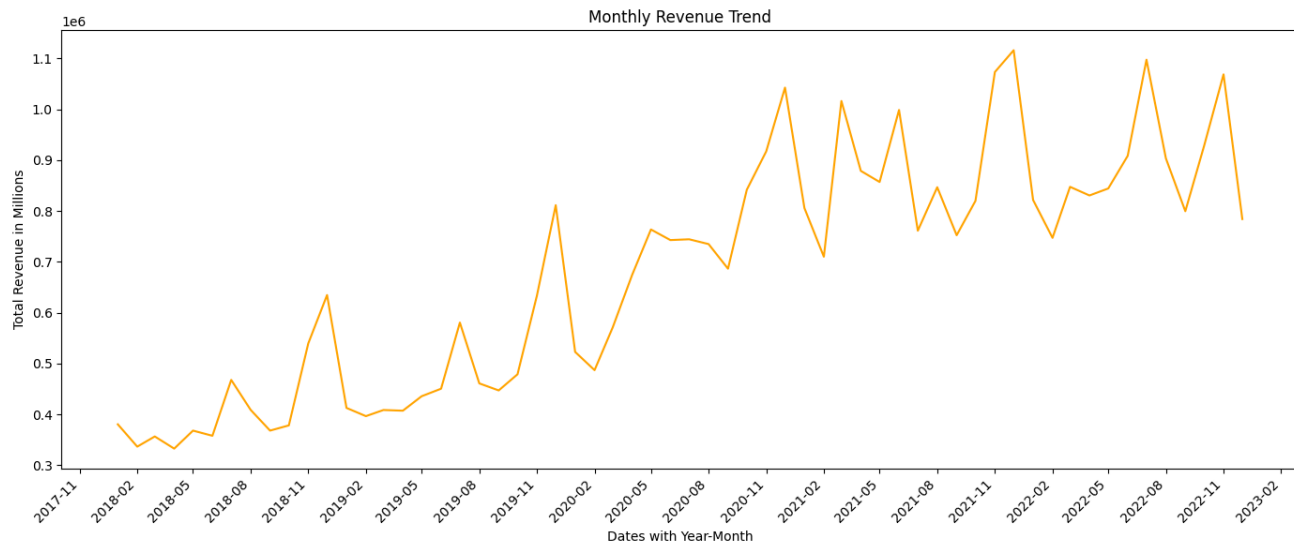


Figure 6 - Monthly purchase-record value trend, 2018–2022

Monthly revenue varies across the observation period, with a pronounced rise around the end of 2021. At weekday level, Monday produces the highest observed weekday-level purchase-record value. These patterns can guide campaign and capacity timing, but they require validation using a complete and representative operational dataset.

3.6 Part A business implications

- Maintain separate tactics for high-volume staple categories and the small share of high-value transactions.
- Use product and state rankings for inventory and localization hypotheses, then normalize by customer or order counts before committing resources.
- Test campaign timing around high-activity periods instead of assuming that historical calendar patterns will persist.
- Treat the descriptive dashboard as a monitoring and hypothesis-generation tool, not as proof of causality.

4. Part B Research Design

4.1 Part B objective

Part B¹² is a separate predictive exercise rather than an integrated continuation of Part A. Its objective is to estimate whether a customer will enter the future *high-value* segment. Historical purchase activity becomes the predictor set, while later spending becomes the outcome. The rationale is practical: if future high-value customers can be identified early, retention and targeting resources can be allocated more efficiently.

4.2 Temporal prediction design

Component	Period / definition	Purpose
Observation window	1 January 2018-31 December 2020	Construct all predictive features
Outcome window	1 January 2021-31 December 2022	Measure future spending only
High-value rule	Future spending $\geq$ \$5,908.16	Label top 25% of future spenders
Unit	One eligible Survey ResponseID	Prevent transaction-level leakage

To prevent data leakage, specifically the principal leakage error where information from the outcome period improperly influences prediction, the `Future_Spending` variable is utilized solely for the construction of the `Target_HighValue` variable. Consequently, `Future_Spending` is rigorously excluded from the feature set `X` used for model training. Customers with no outcome-period purchases receive future spending of zero.

4.3 Target distribution

Class	Customers	Share
Not high value (0)	3,661	74.99%
High value (1)	1,221	25.01%
Total	4,882	100.00%

Because approximately 75% of customers are in the non-high-value class, a majority-class baseline can achieve approximately 75% accuracy while identifying no high-value customers. This is why Part B emphasizes ROC-AUC, precision, recall, and F1-score rather than accuracy alone.

¹² Wibisono, I. A., Agarwal, B., & Schander, R. (2026). *Project B* [Google Colab notebook]. Google Colab. <https://colab.research.google.com/drive/1atpBp7PEz7Ntv10QSUHvwBV5ryDidrn-?usp=sharing>

5. Part B - Predictive Analytics

5.1 Feature engineering

Feature family	Examples	Business meaning
RFM	Recency, Frequency, Monetary	How recently, often, and much the customer purchased
Basket/value	Average, median, maximum purchase value; quantity	Typical and extreme transaction behaviour
Engagement	Active months, active days, purchases per active month	Persistence and purchasing intensity
Breadth/stability	Category diversity, tenure, purchase regularity	Breadth of demand and behavioural consistency
Demographic/context	Age, income, gender, education, state, use frequency	Supplementary customer context

5.2 Preprocessing pipeline

Numerical features are imputed and standardized for scale-sensitive models, while categorical features are imputed with an "Unknown" category and one-hot encoded. All preprocessing steps and the classifier are combined into a scikit-learn pipeline¹³, ensuring transformations are fitted strictly on training data. Using an 80/20 stratified split with a seed of 42 for randomness (`random_state=42`), the dataset is partitioned into a training set of 3,905 customers and a test set of 977 customers.

5.3 Candidate models and evaluation

Model	Reason for inclusion
Dummy baseline	Shows the performance of predicting only the majority class
Logistic Regression	Transparent linear probability benchmark
Decision Tree	Interpretable nonlinear rules
Random Forest	Bagged nonlinear ensemble with interactions
Gradient Boosting	Sequential ensemble designed to reduce residual classification error

ROC-AUC measures ranking quality across thresholds. Precision measures how many predicted high-value customers are truly high value. Recall measures how many actual high-value customers are found. F1 balances precision and recall. Accuracy remains reported but is not sufficient under the 75/25 class distribution.

¹³ Pedregosa, F., et al. (2011). Scikit-learn: Machine Learning in Python. Journal of Machine Learning Research, 12, 2825-2830.

5.4 Untuned model comparison

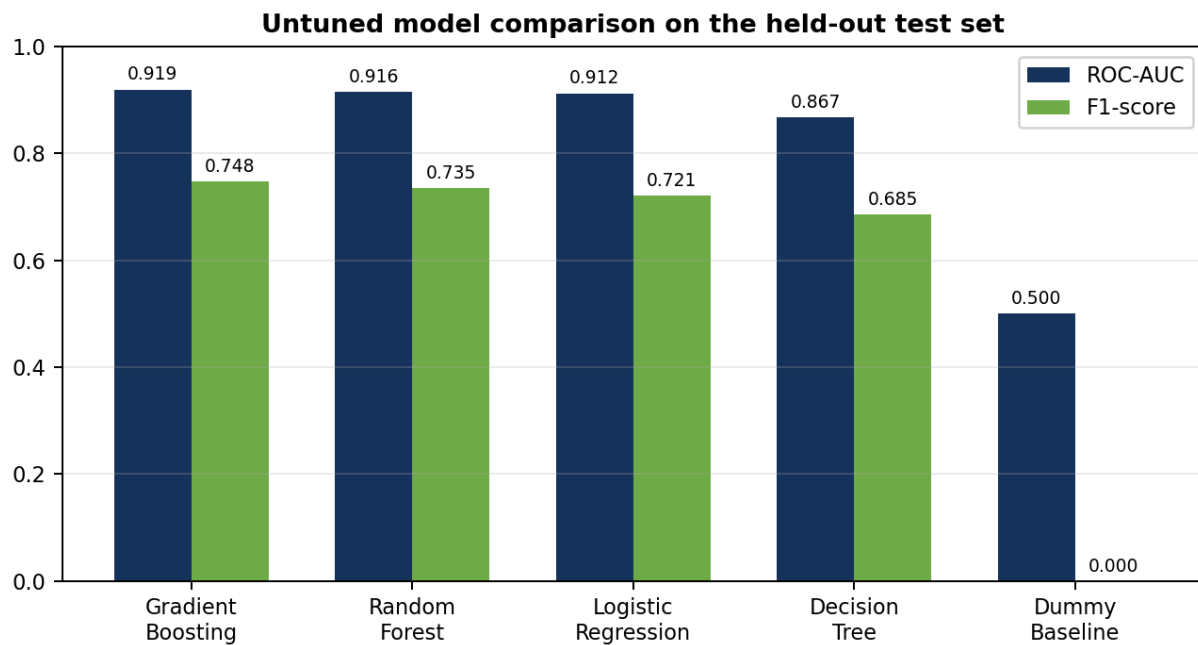


Figure 7 - ROC-AUC and F1-score for the five initial models

Model	Accuracy	Precision	Recall	F1	ROC-AUC
Gradient Boosting	0.879	0.781	0.717	0.748	0.919
Random Forest	0.875	0.782	0.693	0.735	0.916
Logistic Regression	0.870	0.777	0.672	0.721	0.912
Decision Tree	0.847	0.707	0.664	0.685	0.867
Dummy baseline	0.750	0.000	0.000	0.000	0.500

Gradient Boosting achieves the highest ROC-AUC and F1-score. Random Forest has marginally higher precision, but lower recall and F1. Logistic Regression remains competitive, which indicates that much of the predictive signal is systematic rather than exclusively nonlinear. The single Decision Tree is less stable and accurate, while the dummy baseline confirms that raw accuracy can be misleading.

5.5 Feature-set experiment

To compare feature sets fairly, we used the same Random Forest model specification for behavioural-only, demographic-only, and combined inputs. Holding the algorithm constant means that differences in ROC-AUC can be attributed to the information available in each feature set rather than to differences between modelling algorithms.

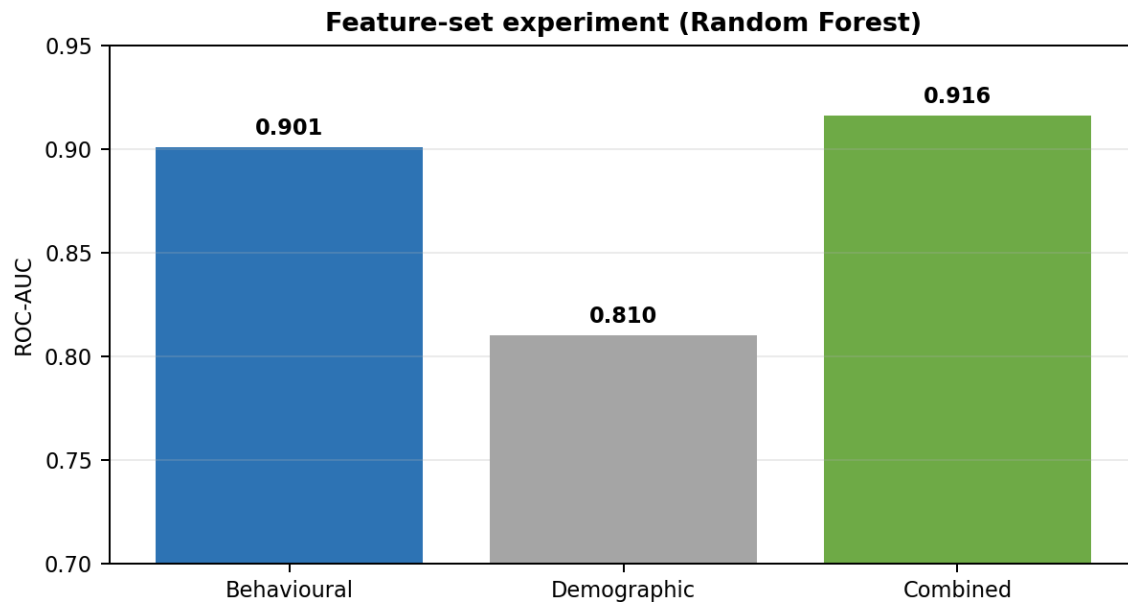


Figure 8 - Predictive contribution of behavioural and demographic feature sets.

With the same Random Forest specification, behavioural features alone achieved ROC-AUC 0.901, compared with 0.810 for demographic features alone. The combined feature set achieved ROC-AUC 0.916.

We used Random Forest to keep the algorithm fixed while comparing the information content of different feature sets. The results show that, within this Random Forest experiment, behaviour is much more predictive than demographics alone, while the combination is best; final model selection is based separately on the stronger Gradient Boosting model comparison and tuning results.

5.6 Hyperparameter tuning

Gradient Boosting is tuned using GridSearchCV with stratified cross-validation on the training data only. The search selects 100 estimators, learning rate 0.1, maximum tree depth 2, and minimum leaf size 10. The best cross-validated training ROC-AUC is 0.924. Keeping the test set untouched until this point protects the final estimate from direct tuning bias.

Hyperparameter	Selected value	Interpretation
n_estimators	100	Number of sequential boosting stages
learning_rate	0.1	Contribution of each new tree
max_depth	2	Shallow weak learners reduce overfitting
min_samples_leaf	10	Regularizes terminal leaves

Model	Accuracy	Precision	Recall	F1-score	ROC-AUC
Gradient Boosting baseline	0.879	0.781	0.717	0.748	0.919
Gradient Boosting tuned	0.874	0.764	0.717	0.740	0.920

Tuning improved ROC-AUC only marginally from 0.919 to 0.920, while accuracy, precision, and F1-score decreased slightly. The selected objective was ROC-AUC, and tuning produced a small improvement.

5.7 Final test-set evaluation

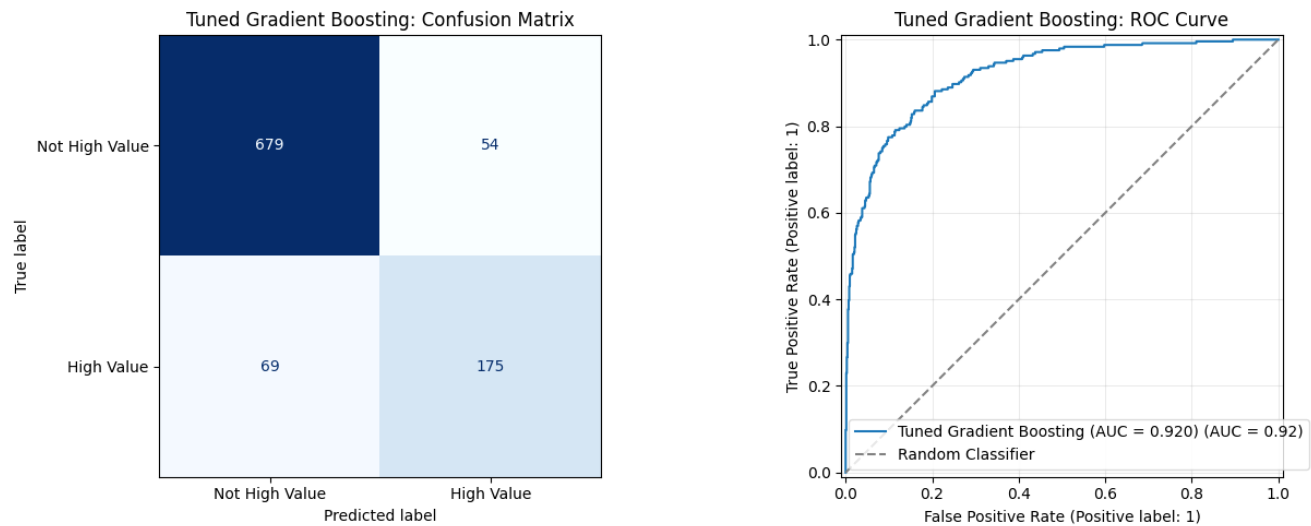


Figure 9 - Final classification outcomes at the 0.50 decision threshold and the ROC curve

Metric	Tuned model result
Accuracy	0.874 (854/977 correctly classified)
High-value precision	0.764
High-value recall	0.717
High-value F1-score	0.740
ROC-AUC	0.920

The final confusion matrix contains 679 true negatives, 175 true positives, 54 false positives, and 69 false negatives. In campaign terms, approximately three out of four customers selected as high value are correct, while the model finds roughly 72% of all actual future high-value customers. The appropriate operating threshold may differ from 0.50: a retention campaign with low contact cost may favour higher recall, whereas an expensive premium-benefit program may favour higher precision.

To evaluate whether the model can prioritise a limited campaign audience, we divided the untouched test-set customers into ten equal-sized groups based on predicted high-value probability; Figure 10 shows the observed high-value rate in each group.

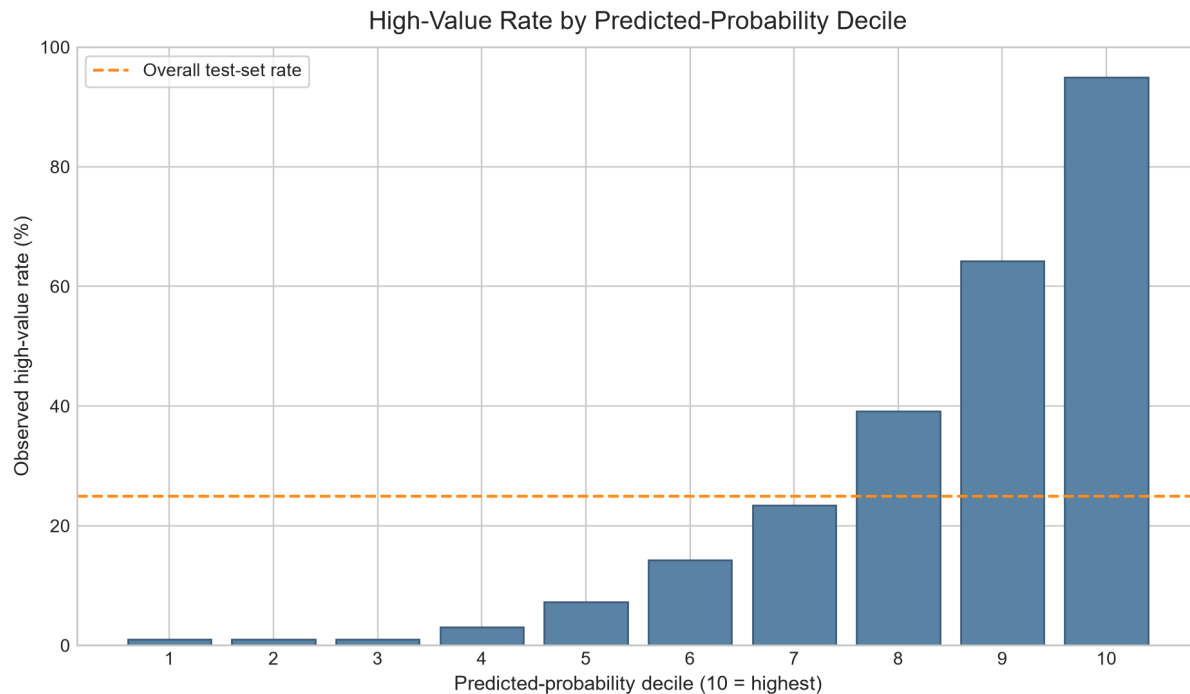


Figure 10 - Observed High-value rate by predicted-probability decile

Figure 10 shows that the observed high-value rate rises sharply as predicted probability increases. The highest-probability decile contains 98 customers, 93 of whom are truly high value, producing a 94.90% high-value rate compared with 24.97% across the full test set. This represents 3.80× lift; the top decile captures 38.11% of all actual high-value customers, while the top two deciles capture 63.93%.

This result supports probability-based campaign prioritisation: rather than contacting all customers, the business can target the highest-ranked group when campaign capacity is limited. The decile profiles in Appendix B are descriptive context only and should not be treated as standalone demographic targeting rules.

5.8 Model interpretation

Signal	Baseline tree importance	Interpretation
Historical Monetary	0.439	Dominant measure of established customer value
Category Diversity	0.257	Broad purchasing across categories predicts continued value
Purchases per Active Month	0.096	Intensity within active periods matters
Use frequency: <5 times/month	0.074	Low stated use helps distinguish lower-value customers
Recency	0.030	Recent activity improves separation of future-value classes

The baseline Gradient Boosting impurity-based importance ranking identifies Monetary and Category Diversity as dominant variables. The final tuned-model permutation analysis provides the primary original-variable interpretation: Monetary is strongest, followed by Amazon-use frequency, Purchases per

Active Month, Recency, and Category Diversity. These rankings are interpreted as predictive signals rather than causal effects.

5.9 Business translation

- Retention: prioritize customers with high historical monetary value for tiered loyalty benefits and service protection.
- Cross-selling: use category diversity to identify receptive customers and introduce adjacent categories, while testing relevance and margin.
- Triggered engagement: contact customers during their observed active cycles and monitor whether frequency increases without excessive discounting.
- Win-back: target customers with long recency or low stated platform use using lower-cost reactivation offers.
- Capacity-based ranking: use predicted probabilities to fill a fixed campaign audience rather than treating the binary class as an unquestionable decision.

5.10 Requirement coverage

Part B requirement	Evidence in the project
Preprocessing	Customer aggregation, imputation, scaling, one-hot encoding, leakage controls
Suitable models	Dummy, Logistic Regression, Decision Tree, Random Forest, Gradient Boosting
Training and evaluation	Stratified 80/20 split; accuracy, precision, recall, F1, ROC-AUC
Hyperparameter tuning	GridSearchCV with stratified cross-validation
Interpretation	Tree importance and tuned-model permutation importance
Business strategy	Retention, cross-selling, activity triggers, win-back, probability-based targeting

6. Part B Business Recommendations

6.1 Recommended operating model

Priority	Action	Evidence	Control
1	Protect established high spenders	Monetary is the strongest predictive signal	Use incremental holdout tests; avoid blanket discounts
2	Expand category breadth	Category diversity is the second strongest baseline signal	Optimize for contribution margin, not clicks alone
3	Time engagement to active cycles	Purchases per active month is highly informative	Frequency caps and channel-consent rules
4	Re-engage dormant customers	Recency separates future-value outcomes	Limit spend when predicted value is low
5	Localize merchandising hypotheses	Part A reveals category and state differences	Normalize by market size and validate out of sample

6.2 Campaign threshold framework

The model should produce a probability score for each customer. A business threshold should then be chosen using campaign economics. If contact cost is low and missing a valuable customer is expensive, lower the threshold to increase recall. If incentives are costly or exclusive, raise the threshold to increase precision. The final decision rule should be documented before results are reviewed and tested against a randomly selected control group.

6.3 Measurement plan

- Primary commercial metric: incremental contribution margin per targeted customer.
- Model metrics: ROC-AUC, precision, recall, F1, calibration, and lift in the selected audience.
- Operational metrics: audience size, delivery rate, redemption, frequency, and offer cost.
- Guardrails: unsubscribe rate, complaint rate, demographic disparities, and excessive discount dependence.

6.4 Deployment sequence

- Score an eligible customer population using only information available at the scoring date.
- Rank customers by predicted probability and select a threshold based on budget and campaign capacity.
- Randomly hold out a control group within the targeted range.
- Run differentiated retention, cross-selling, or re-engagement treatments.
- Measure incremental value rather than raw post-campaign revenue.
- Monitor model drift and retrain only after documenting changes in data coverage and business policy.

7. Conclusion

Part A shows that observed Amazon revenue in this sample is concentrated across transaction values, categories, customer segments, geography, and time. Its main contribution is descriptive: it identifies where value appears in the historical data and clarifies several limitations around representation, category aggregation, and interpretation.

Part B demonstrates that future high-value status can be predicted with useful ranking quality from earlier behaviour and selected customer characteristics. Gradient Boosting is the strongest tested model, reaching ROC-AUC 0.920 after tuning via cross-validation on the training data and then being evaluated once on the untouched final test set. Behavioural data carries most of the signal; demographics provide a smaller incremental contribution. At the default threshold, the model balances precision and recall sufficiently to support targeted experiments, and the probability-decile results show that the highest-ranked customers are strongly enriched for future high-value outcomes. Even so, the model should not be deployed without cost-sensitive threshold selection, fairness review, calibration checks, ranked-audience validation, and controlled measurement of incremental commercial impact.

Taken together, the document reports one descriptive task and one predictive task on the same dataset, but each should be evaluated on its own terms. Part A is useful for sample-level insight and hypothesis generation. Part B is useful for ranked targeting and controlled experimentation. The next analytical step is not necessarily a more complex algorithm, but a better decision experiment that tests whether model-guided targeting creates incremental business value.

8. Code Availability and Reproducibility

The analysis was implemented in Google Colab using Python. The notebooks contain the complete data preparation, exploratory analysis, feature engineering, predictive modelling, evaluation, visualisations, and dashboard code used for this report.

Part A – Descriptive Analysis and Exploratory Data Analysis:

https://colab.research.google.com/drive/1LYEbNHwc8IXNQG5J9GoL0Vv_dTkzWcAc?usp=sharing

Part B – Predictive Analysis and High-Value Customer Classification:

<https://colab.research.google.com/drive/1atpBp7PEz7Ntv10QSUHvwBV5ryDidrn-?usp=sharing>

To reproduce the analysis, place amazon-purchases.csv, survey.csv, and the category-mapping file in the specified Google Drive folder structure before running the notebooks from top to bottom.

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Appendix A - Detailed Part A Rankings

Rank	Original category	Revenue (US\$)
1	ABIS_BOOK	1,337,049.86
2	GIFT_CARD	887,639.45
3	PET_FOOD	752,841.51
4	NUTRITIONAL_SUPPLEMENT	604,105.18
5	NOTEBOOK_COMPUTER	578,418.88
6	SHOES	562,050.29
7	HEADPHONES	535,391.15
8	SHIRT	504,363.13
9	PANTS	397,926.69
10	HEALTH_PERSONAL_CARE	308,238.83

Rank	State	Revenue (US\$)
1	CA	4,407,187.16
2	TX	3,139,704.59
3	FL	2,604,459.28
4	NY	2,293,724.09
5	PA	2,207,400.75
6	OH	1,880,774.12
7	IL	1,721,189.01
8	NC	1,652,396.09
9	MI	1,330,336.06
10	GA	1,298,669.47

Appendix B - Predictive Feature Dictionary

Appendix Figure: Descriptive profile of customers by predicted-probability decile.

Descriptive Profile by Predicted-Probability Decile

Decile	Q-demos-age	Q-demos-income	Q-demos-gender	Q-demos-education	Q-demos-state	Q-amazon-use-how-oft
9	10	35 - 44 years (28.6%)	\$100,000 - \$149,999 (34.7%)	Female (56.1%)	Bachelor's degree (44.9%)	California (10.2%)
8	9	25 - 34 years (31.6%)	\$100,000 - \$149,999 (27.6%)	Male (53.1%)	Bachelor's degree (40.8%)	California (14.3%)
7	8	25 - 34 years (39.2%)	\$50,000 - \$74,999 (27.8%)	Female (51.5%)	High school diploma or GED (41.2%)	Texas (12.4%)
6	7	25 - 34 years (40.8%)	\$25,000 - \$49,999 (21.4%)	Female (50.0%)	Bachelor's degree (44.9%)	California (9.2%)
5	6	25 - 34 years (37.8%)	\$25,000 - \$49,999 (23.5%)	Female (50.0%)	Bachelor's degree (43.9%)	New York (8.2%)
4	5	25 - 34 years (30.9%)	\$50,000 - \$74,999 (29.9%)	Female (54.6%)	Bachelor's degree (47.4%)	Florida (8.2%)
3	4	25 - 34 years (40.8%)	\$25,000 - \$49,999 (29.6%)	Female (48.0%)	Bachelor's degree (40.8%)	California (12.2%)
2	3	25 - 34 years (44.3%)	\$25,000 - \$49,999 (24.7%)	Male (49.5%)	Bachelor's degree (48.5%)	California (9.3%)
1	2	25 - 34 years (32.7%)	\$25,000 - \$49,999 (31.6%)	Female (49.0%)	Bachelor's degree (49.0%)	California (9.2%)
0	1	25 - 34 years (39.8%)	\$50,000 - \$74,999 (26.5%)	Female (54.1%)	Bachelor's degree (45.9%)	New York (9.2%)

Feature	Definition
Recency	Days between the customer's last historical purchase and the observation-window endpoint.
Frequency	Number of historical purchase records.
Monetary	Historical purchase-record value.
Avg / Median / Max Purchase Value	Distribution of historical line-item values for the customer.
Total Quantity	Sum of purchased units in the historical window.
Active Purchase Months / Days	Number of distinct calendar months or days with purchases.
Category Diversity	Number of distinct product categories purchased.
Customer Tenure	Elapsed time between first and last observed historical activity.
Purchase Regularity	Consistency or dispersion of time gaps between purchases.
Purchases per Active Month	Frequency normalized by the number of active months.
Selected demographics	Age, income, gender, education, state, and reported Amazon-use frequency.